UNIVERSITY
OF WOLLONGONG
IN DUBAI

Individual Lab Report Cover Sheet

Complete all sections of this cover sheet before submitting.

Student name:	Subiksha Sri Gunaseelan
Student number:	1126544
Subject number and name:	CSCI291 – Programming for Engineering
Subject coordinator:	Dr. Obada Al Khatib
Title of assignment:	Lab Project 1 – Automated Car Wash and Detailing Bay Controller
Lab number:	4
Tutor / lab instructor name:	Eng. Yasmine Abualroos
Date submitted:	16 October 2026
Total number of pages:	10

Student Declaration and Acknowledgement

By submitting this assignment, I declare that:

1. I have read the subject outline for this subject, and this assessment item meets the requirements detailed therein.

2. This assessment is entirely my own work, except where I have included fully documented references to the work of others. The material contained in this assessment item has not previously been submitted for assessment in this or any other subject.

3. Acknowledgement of source information is in accordance with the guidelines or referencing style specified in the subject outline.

4. I am aware of the late submission policy and penalty.

5. I either did not use any Artificial Intelligence (AI) tools in the completion of this assessment, or where I did use AI tools, this use was disclosed to and approved by my instructor prior to submission.

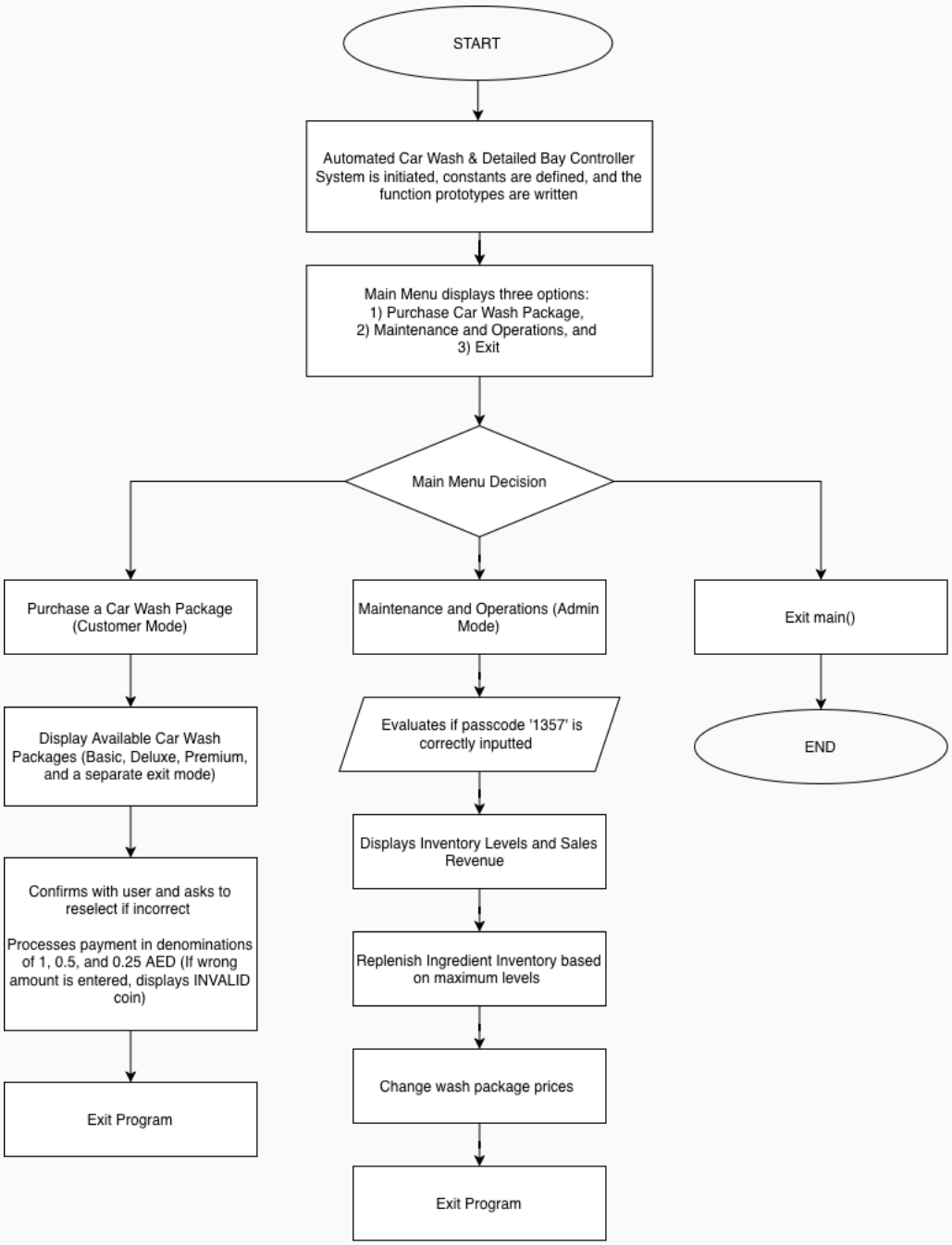
6. I understand that undisclosed or unapproved use of AI tools in this assessment constitutes academic misconduct and may result in penalties up to and including a mark of zero, in accordance with UOWD's Academic Integrity Policy.

Signature:		**Date:**	16 October 2026

Section 1 — Pre-Lab Design (5 marks)

Instructions: *Insert your two pre-lab diagrams below. These should be the same diagrams you brought to Week 2 and had signed off, redrawn neatly, or photographed clearly.*

1.1 Function Hierarchy Chart



Caption: Figure 1.1 — Function hierarchy chart showing main() and it's called functions.

1.2 Payment Flowchart



Caption: Figure 1.2 — Flowchart of the coin-insertion payment process.

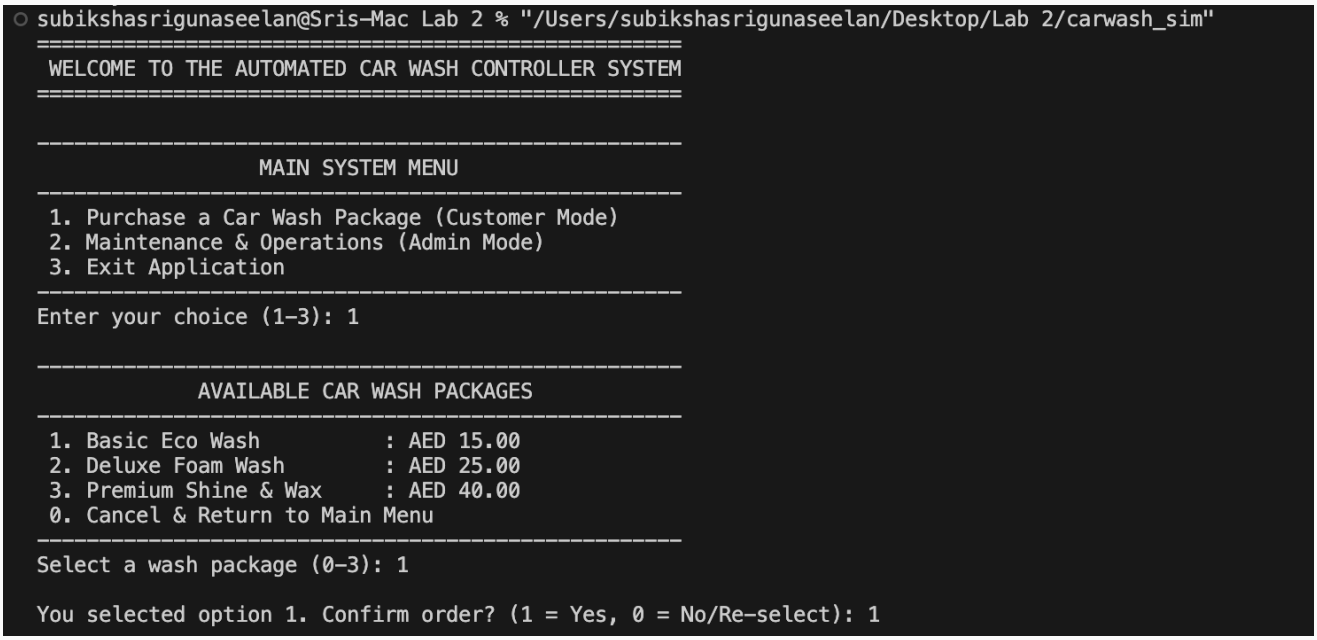
Section 2 — Main Architecture & Menu System

Instructions: *Briefly describe your code, then insert a screenshot of the running main menu.*

2.1 Description

The code is for an Automated Car Wash and Detailing Bay Controller. The Main System Menu has 3 options: Customer Mode (where a package can be purchased), Admin Mode (where maintenance and operations can be completed) and Exit Application to suspend operations and stop the program from running. The Customer Mode allows the user to select from three wash packages, confirm the order, and proceed with payment. It also displays an alert if there are low supplies before selection of the package and a low stock alert is triggered after payment. The Admin Mode checks for a set password and enables access to display inventory levels and sales revenue, replenish ingredient inventory, change wash package prices, and reset total sales revenue.

2.2 Screenshot — Main Menu



Caption: Figure 2.1 — Main menu on program startup.

Section 3 — Customer Mode & Payment Loop

Instructions: *Insert one screenshot for each test case below, showing the terminal output. Make sure the full relevant output is visible in each screenshot.*

3.1 Successful Purchase with Exact/Over Payment

```
==================================================
PAYMENT PROCESS: Basic Eco Wash
TOTAL AMOUNT DUE: AED 15.00
Accepted Coins : 1.00 AED, 0.50 AED, 0.25 AED
==================================================

Outstanding Balance : AED 15.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 1.00

Outstanding Balance : AED 14.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 2.00

Outstanding Balance : AED 13.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 3.00

Outstanding Balance : AED 12.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 4.00

Outstanding Balance : AED 11.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 5.00
```

```
Outstanding Balance : AED 10.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 6.00

Outstanding Balance : AED 9.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 7.00

Outstanding Balance : AED 8.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 8.00

Outstanding Balance : AED 7.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 9.00

Outstanding Balance : AED 6.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 10.00

Outstanding Balance : AED 5.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 11.00
```

```
Outstanding Balance : AED 4.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 12.00

Outstanding Balance : AED 3.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 13.00

Outstanding Balance : AED 2.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 14.00

Outstanding Balance : AED 1.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 15.00


==================================================
SUCCESS! DISPENSING WASH SERVICE: Basic Eco Wash
--------------------------------------------------
Package Price : AED 15.00
Total Paid    : AED 15.00
Change Due    : AED 0.00
==================================================
Please drive your vehicle into the bay. Thank you!
```

Caption: Figure 3.1 — Successful purchase, coins accepted, correct change shown.

3.2 Package Unavailable (Low Supplies)

```
subikshasrigunaseelan@Sris-Mac Lab 2 % "/Users/subikshasrigunaseelan/Desktop/Lab 2/carwash_sim"
==================================================
 WELCOME TO THE AUTOMATED CAR WASH CONTROLLER SYSTEM
==================================================

--------------------------------------------------
                MAIN SYSTEM MENU
--------------------------------------------------
 1. Purchase a Car Wash Package (Customer Mode)
 2. Maintenance & Operations (Admin Mode)
 3. Exit Application
--------------------------------------------------
Enter your choice (1-3): 1

--------------------------------------------------
            AVAILABLE CAR WASH PACKAGES
--------------------------------------------------
 1. Basic Eco Wash          : AED 15.00
 2. Deluxe Foam Wash        : AED 25.00 [UNAVAILABLE - Low Supplies]
 3. Premium Shine & Wax     : AED 40.00 [UNAVAILABLE - Low Supplies]
 0. Cancel & Return to Main Menu
--------------------------------------------------
Select a wash package (0-3): 2
[UNAVAILABLE - Low Supplies] Please select another package.

--------------------------------------------------
            AVAILABLE CAR WASH PACKAGES
--------------------------------------------------
 1. Basic Eco Wash          : AED 15.00
 2. Deluxe Foam Wash        : AED 25.00 [UNAVAILABLE - Low Supplies]
 3. Premium Shine & Wax     : AED 40.00 [UNAVAILABLE - Low Supplies]
 0. Cancel & Return to Main Menu
--------------------------------------------------
```

Caption: Figure 3.2 — Package correctly flagged unavailable when resources are insufficient.

3.3 Invalid Coin Rejected

```
Outstanding Balance : AED 14.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 2.00

Outstanding Balance : AED 13.00
Insert Coin (1.0, 0.50, 0.25): 2
[REJECTED] Invalid coin (AED 2.00). Please collect your coin.

Outstanding Balance : AED 13.00
Insert Coin (1.0, 0.50, 0.25): 2
[REJECTED] Invalid coin (AED 2.00). Please collect your coin.

Outstanding Balance : AED 13.00
Insert Coin (1.0, 0.50, 0.25): 2
[REJECTED] Invalid coin (AED 2.00). Please collect your coin.

Outstanding Balance : AED 13.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 3.00
```

Caption: Figure 3.3 — Invalid coin rejected; valid coins still accepted afterward.

A successful purchase is defined after the price of the package is paid in denominations of 1.00 AED, 0.50 AED, or 0.25 AED. Any amount inserted more than this is rejected, and the program prompts the user to collect the coin and insert the correct amount again till the price is paid completely.

3.4 Low-Stock Alert Triggered

```
Outstanding Balance : AED 3.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 13.00

Outstanding Balance : AED 2.00
Insert Coin (1.0, 0.50, 0.25): 1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 14.00

Outstanding Balance : AED 1.00
Insert Coin (1.0, 0.50, 0.25):
1
[ACCEPTED] Coin AED 1.00 added. Total Paid: AED 15.00


==================================================
SUCCESS! DISPENSING WASH SERVICE: Basic Eco Wash
--------------------------------------------------
Package Price : AED 15.00
Total Paid    : AED 15.00
Change Due    : AED 0.00
==================================================
Please drive your vehicle into the bay. Thank you!

[OPERATOR ALERT] Water supply is low: 30.0 L

[OPERATOR ALERT] Soap concentrate is low: 200.0 mL
```

Caption: Figure 3.4 — Low-stock alert printed after inventory drops below threshold.

After the payment is completed and the wash service is dispensed, an operator alert is displayed to inform about low supply levels in the machine and the package will be unavailable in the forthcoming iterations.

Section 4 — Admin Operations Mode (10 marks)

Instructions: *Insert one screenshot for each test case below.*

4.1 Incorrect Passcode Denied

```
------------------------------------------------------
                  MAIN SYSTEM MENU
------------------------------------------------------
 1. Purchase a Car Wash Package (Customer Mode)
 2. Maintenance & Operations (Admin Mode)
 3. Exit Application
------------------------------------------------------
Enter your choice (1-3): 2

------------------------------------------------------
              ADMIN MAINTENANCE MODE ACCESS
------------------------------------------------------
Enter Admin Passcode: 1234

[ACCESS DENIED] Incorrect passcode entered!

------------------------------------------------------
                  MAIN SYSTEM MENU
------------------------------------------------------
```

Caption: Figure 4.1 — Incorrect passcode correctly denies access.

4.2 Correct Passcode + Inventory & Sales Display

```
------------------------------------------------------
              ADMIN MAINTENANCE MODE ACCESS
------------------------------------------------------
Enter Admin Passcode: 1357

[ACCESS GRANTED] Welcome, System Operator.

------------------------------------------------------
                 ADMIN OPERATIONS MENU
------------------------------------------------------
 1. Display Inventory Levels & Sales Revenue
 2. Replenish Ingredient Inventory
 3. Change Wash Package Prices
 4. Reset Total Sales Revenue
 0. Exit Admin Mode
------------------------------------------------------
Select an admin operation (0-4): 1

------------------------------------------------------
          INVENTORY LEVELS & SALES REVENUE
------------------------------------------------------
Current Water: 10.0 L (Max: 300.0 L)
Current Soap: 100.0 mL (Max: 2000.0 mL)
Current Wax: 400.0 mL (Max: 1000.0 mL)
------------------------------------------------------
Basic Eco Wash Price       : AED 15.00
Deluxe Foam Wash Price     : AED 25.00
Premium Shine & Wax Price  : AED 40.00
------------------------------------------------------
Total Sales Revenue        : AED 105.00
======================================================
```

Caption: Figure 4.2 — Admin access granted; inventory and sales report displayed.

4.3 Replenish Ingredients (Max Capacity Check)

```
------------------------------------------------------
                 ADMIN OPERATIONS MENU
------------------------------------------------------
 1. Display Inventory Levels & Sales Revenue
 2. Replenish Ingredient Inventory
 3. Change Wash Package Prices
 4. Reset Total Sales Revenue
 0. Exit Admin Mode
------------------------------------------------------
Select an admin operation (0-4): 2

------------------------------------------------------
              REPLENISH INGREDIENT INVENTORY
------------------------------------------------------
Current Water: 10.0 L (Max: 300.0 L)
Enter Water to add (Liters): 300
Current Soap: 100.0 mL (Max: 2000.0 mL)
Enter Soap to add (mL): 2000
Current Wax: 400.0 mL (Max: 1000.0 mL)
Enter Wax to add (mL): 1100

======================================================
 [SUCCESS] INVENTORY REPLENISHED SUCCESSFULLY!
------------------------------------------------------
 Updated Water Level : 300.0 L
 Updated Soap Level  : 2000.0 mL
 Updated Wax Level   : 1000.0 mL
======================================================
```

Caption: Figure 4.3 — Replenishment capped at maximum tank capacity, with notice shown.

4.4 Change Package Price

```
------------------------------------------------------
                 ADMIN OPERATIONS MENU
------------------------------------------------------
 1. Display Inventory Levels & Sales Revenue
 2. Replenish Ingredient Inventory
 3. Change Wash Package Prices
 4. Reset Total Sales Revenue
 0. Exit Admin Mode
------------------------------------------------------
Select an admin operation (0-4): 3

------------------------------------------------------
              CHANGE WASH PACKAGE PRICES
------------------------------------------------------
1. Basic Eco Wash       : AED 15.00
2. Deluxe Foam Wash     : AED 25.00
3. Premium Shine & Wax  : AED 40.00
0. Return to Admin Menu
------------------------------------------------------
Select a package to modify (0-3): 1
Enter new price (AED): 16.7

[SUCCESS] Wash package price updated successfully!

------------------------------------------------------
              CHANGE WASH PACKAGE PRICES
------------------------------------------------------
1. Basic Eco Wash       : AED 16.70
2. Deluxe Foam Wash     : AED 25.00
3. Premium Shine & Wax  : AED 40.00
0. Return to Admin Menu
------------------------------------------------------
```

Caption: Figure 4.4 — Package price successfully updated.

4.5 Reset Sales Revenue

```
--------------------------------------------------
                ADMIN OPERATIONS MENU
--------------------------------------------------
 1. Display Inventory Levels & Sales Revenue
 2. Replenish Ingredient Inventory
 3. Change Wash Package Prices
 4. Reset Total Sales Revenue
 0. Exit Admin Mode
--------------------------------------------------
Select an admin operation (0-4): 4

--------------------------------------------------
                RESET TOTAL SALES REVENUE
--------------------------------------------------
Current Sales Revenue: AED 105.00
Please collect the cash box earnings before resetting.
Confirm revenue reset (1 = Yes, 0 = No): 1

[SUCCESS] Total sales revenue reset to AED 0.00.
```

```
--------------------------------------------------
                ADMIN OPERATIONS MENU
--------------------------------------------------
 1. Display Inventory Levels & Sales Revenue
 2. Replenish Ingredient Inventory
 3. Change Wash Package Prices
 4. Reset Total Sales Revenue
 0. Exit Admin Mode
--------------------------------------------------
Select an admin operation (0-4): 1

--------------------------------------------------
          INVENTORY LEVELS & SALES REVENUE
--------------------------------------------------
Current Water: 150.0 L (Max: 300.0 L)
Current Soap: 800.0 mL (Max: 2000.0 mL)
Current Wax: 400.0 mL (Max: 1000.0 mL)
--------------------------------------------------
Basic Eco Wash Price       : AED 15.00
Deluxe Foam Wash Price     : AED 25.00
Premium Shine & Wax Price  : AED 40.00
--------------------------------------------------
Total Sales Revenue        : AED 0.00
==================================================
```

Caption: Figure 4.5 — Sales revenue reset after confirmation.

Source Code Submission

Your full, final, compiling carwash_sim.c is submitted as a separate file alongside this report — do not paste the full source code into this document.

File naming: CSCI291_LabProject1_StudentName_ID.c — uploaded together with this report under the LabSubmission Link on Moodle.